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HW Set 06: Bayesian Inference

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ABSTRACT

The fifth homework: it's Bayesian inferencing time. All code can be found in the Jupyter Notebook: https://github.com/kparrine/Computational_Astro/blob/main/C6/C6.ipynb.

1. QUESTION 1

The analytical computation can be found here: https://github.com/kparrine/Computational_Astro/blob/main/C6/Q1.pdf.

2. QUESTION 2

The code for my Bayesian inference is in the link in the abstract, however, I will put my final function here without comments or the docstring:

```
def Bayes(dataname,a,b):
    if isinstance(dataname, str):
        data = np.loadtxt(dataname)
    else:
        data = dataname

    n = len(data)
    h = int(np.sum(data))

    likelihood = make_likelihood(n, h)
    prior = make_prior(a, b)
    posterior = make_posterior(prior, likelihood)

    return posterior
```

Figure 1 shows my probability distribution function for the coin toss similar to Q1a using the provided data in data.txt. The coin is not balanced because there is a higher probability that the coin will land on heads. Figure 2 shows the distribution when 5, 50 and 500 coin flips are used. As the number of coin flips increases the curve gets thinner and the distribution focuses more on a specific value rather than a wide range of potential values.

Figure 3 shows the results for different prior combinations with different amounts of data. In (a), I have a flat prior in which I have no idea if the coin is fair, in (b), I have a weak belief in fairness,

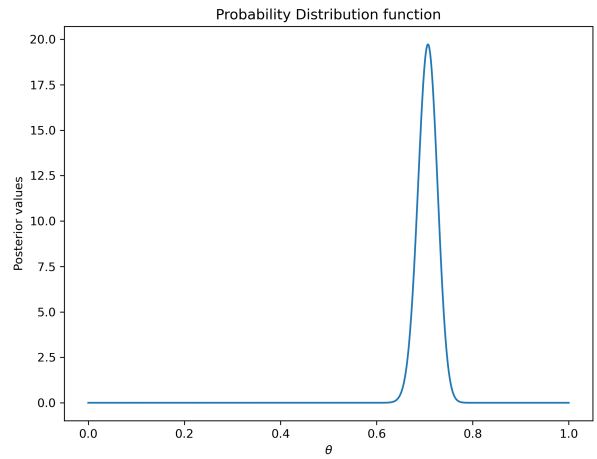

Figure 1. Probability distribution function for the coin flip problem.

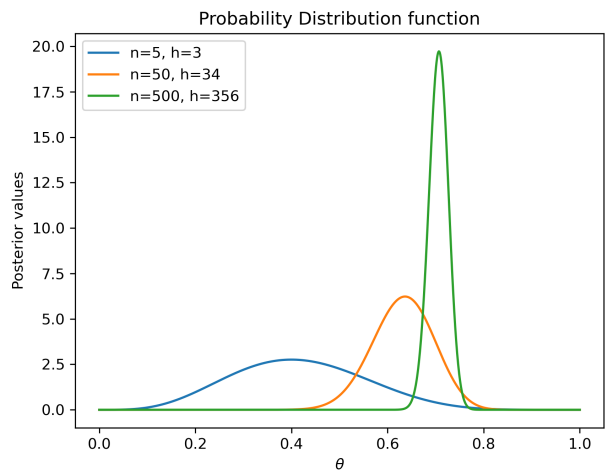

Figure 2. Probability distribution for three different scenarios, in blue is when 5 coin flips are used, in orange is when 50 coin flips are used, and in green is when 500 coin flips are used.

so an openness to be wrong, in (c), I have a roughly fair coin, in (d), I have a strong belief the coin biased towards tails, in (e), I have a strong bias towards heads, and in (f), I have a very strong bias towards heads. In all plots, the curve with the highest number of coin flips looks the same: thin and honing in on a specific value of θ; as if the priors no longer matter. I also notice that when the priors are small, the lower number of coin flips has a higher spread of potential values, whereas, at larger priors, the curves are more thin. Coin flips of 5 and 50 are more affected by my prior values, especially at the lowest number of coin flips. Noticeably in (e), there is a good agreement with the three distributions because the coin is actually biased towards heads, however, not overly so as seen in (f). In this case, as said earlier, a lower number of coin flips affect the priors more, so they are too biased towards heads. The weight of priors with a low number of coin flips is also seen in (d), where I set a prior to be more towards tails even though the actual distribution should be towards heads.

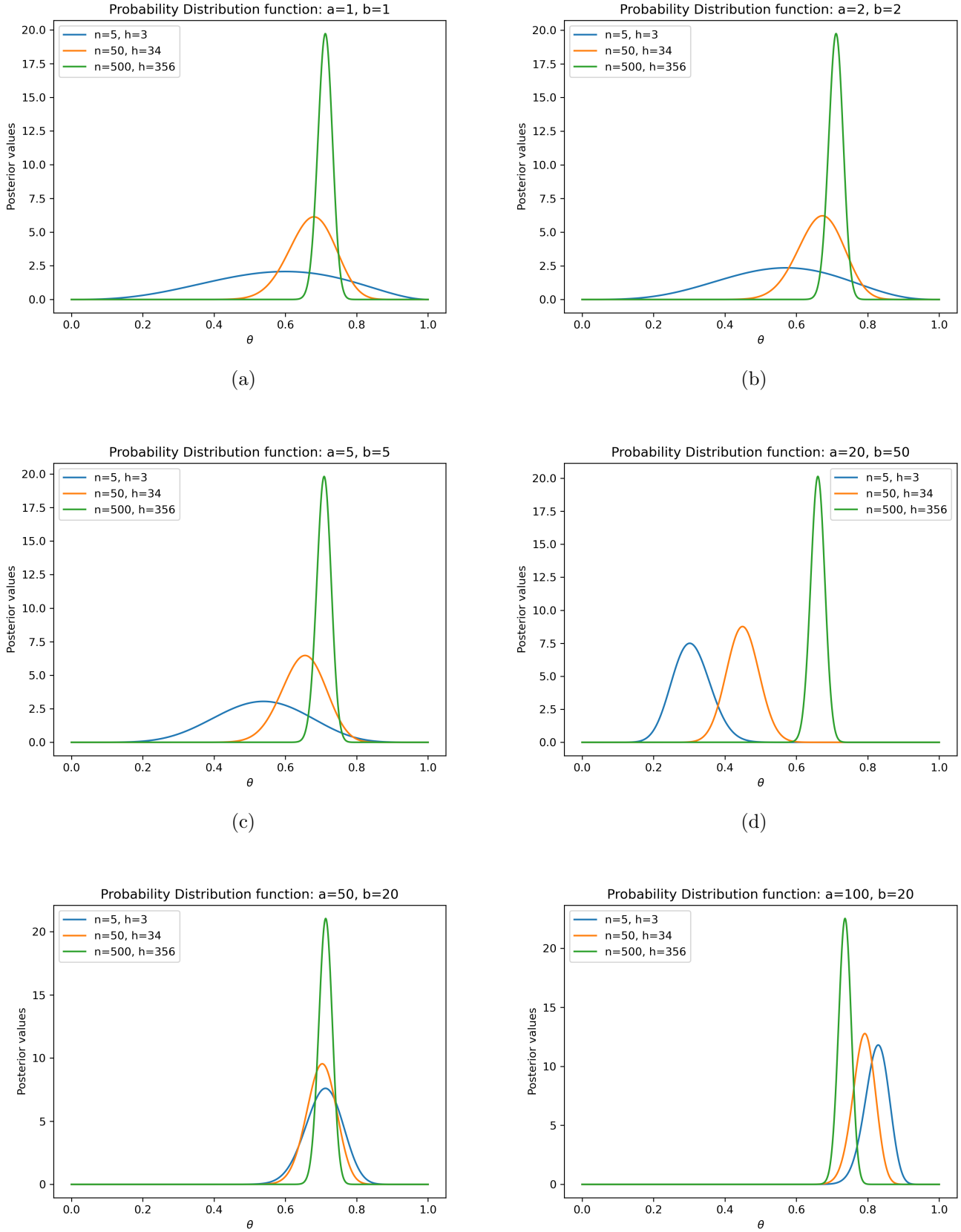

Figure 3. Probability distribution with different priors for three different sets of coin flips, in blue is when 5 coin flips are used, in orange is when 50 coin flips are used, and in green is when 500 coin flips are used. (a) has priors (1, 1), (b) has priors (2, 2), (c) has priors (5, 5), (d) has priors (20, 50), (e) has priors (50, 70), and (f) has priors (100, 20).

Table 1. Calculation of π

N	Priors (a,b)	h	P	π_{calc}	Relative Error (%)
50	(10, 20)	36	0.575	2.3	26.79
50	(3, 3)	42	0.804	3.214	2.31
500	(2, 2)	385	0.768	3.071	2.233
1000	(10, 2)	804	0.804	3.217	2.413
1400	(3, 3)	1124	0.802	3.206	2.058
1400	(10, 15)	1111	0.787	3.147	0.162

3. QUESTION 3

I create my randomized data selection by adhering to the instructions laid out, calculate the number of points inside the circle of radius one defined as h, then calculate the probability of a point falling inside the circle defined as P, then I use use $\pi = \frac{A}{r^2}$ to find a value for π, defined as π_{calc}, finally, I also test how the relative error is affected by the prior and the number of data points defined as N used. Table 1 shows these results.

When I have a low number of data points, the prior matters a lot more because the error gets very large if I don't place my priors well. However, at a large number of data points, my priors do not seem to matter as much because the error stays below 2.5%. The most interesting was my final run of 1400 datapoints because I got a value very close to π.